Course Syllabus

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Medical Microbiology and Immunology (MED602)

Teaching Faculty

Michael D. Powell, PhD	Professor and Course Director	RW336	752-1582
Erica L. Johnson, PhD	Assoc. Professor	RW203	756-6660
Austin Chan, MD	**Assoc Professor**	**Medicine**	**404-756-1315**
Udhayakumar, Venkatachalam	**Adjunct Professor**	**MBI**	
Kenneth Onyedibe, MD, PhD	**Assoc. Professor**		
Jonathan Stiles, PhD	**Professor**	HG352	752-1585
Barney Graham, MD, PhD	**Professor**	bgraham@msm.edu (mailto:bgraham@msm.edu)	

Course Organization

The Medical Microbiology and Immunology course consists of introductory material (Including immunology), a review of infections by organ system, and an overview of bacteria, viruses, and fungal and parasitic organisms.

To give you the required information in as flexible a format as possible we present most of the material as “Self-Study or Online lectures”. Information is presented in pre-recorded presentations that you may review at your own discretion. In most cases, we will have a follow up class period marked “Discussion” immediately afterward to discuss the information presented in pre-recorded lectures and present cases or problems to help ensure mastery of the subject. We employ Turning Point real-time questions to give an example of the topics to expect in follow up examinations.

Specific questions about lecture material should be made directly to the lecturer presenting the lecture. Questions regarding the course and grading in general should be directed to Dr. Powell. In instances where responses prove unsatisfactory the student may appeal to Dr. Elks.

At any time during the course that a student's grade falls below a "C", the student is expected to discuss the problem with the course director. Occasionally tutoring will be arranged for those students requiring additional help. Questions and/or concerns regarding examination questions are to be presented to the class representative who will present these "problem questions" to the course director and/or the faculty representative; this will be done no later than two weeks after the exam is administered.

Specific Objectives :

A certain amount of understanding of cell biology, biochemistry, and genetics mastery are assumed from first year classes. We will often review some of these concepts, but it is assumed that the student comes prepared with an understanding of these subjects. Each year, a large part of understanding Microbiology is dealing with the naming and organization of organisms. Although we will cover these aspects for each organism presented, the student is encouraged to re-enforce these topics with reference to the text when needed.

Section I – what information will we need to know to understand and treat infectious diseases?

The main objectives for this section are:

1. to introduce the organisms that can cause infectious disease
2. to discuss how the body fights off infection by
3. to discuss how the organisms, in turn, can defeat these measures to cause disease
4. to show the common means used to detect microorganisms
5. to discuss common anti-virals and their mechanism of action and discuss antimicrobial susceptibility testing.

In each section, we will ask Turning Point in class questions to test mastery of each topic and give you examples of the type of questions that will be asked on exams.

Section II – Infections of specific organ systems

The main goal of this section is to discuss the organisms that can cause infections of specific organ systems. Included in this discussion are:

1. What are the common pathogens that cause infections specific to each organ-system?
2. What are the outstanding morphologic or replicative strategies used by these organisms?
3. What are the main features of the pathogenesis of disease of these organisms?

***Section III* - Case Studies**

Most lectures will contain examples of the use of the material presented in the lectures as relevant examples. These examples will typically take the form of case studies or vignettes. Students will be expected to be able to synthesize examples of how the principles presented will be used in the diagnosis of typical infections.

More and more the national board exams also test mastery of a subject by the student's ability to interpret experimental data. Some of the case studies will therefore include "experimental" cases where the student can practice their ability to interpret these types of data. By the end of this section the student should be proficient at identifying the key features of each infection and how to distinguish one type of infection from another.

Instructional Materials

a) Lecture PowerPoints:

Each instructor will provide copies of the PowerPoint slides presented during lecture. PowerPoint slides will be made available through the Course Canvas site. Our intent is to provide you the **MOST** relevant material in these PowerPoint presentations. Be aware that often the topics for lecture will often require an understanding of the concepts presented and not simple memorization. The reading assignments provide you an alternative discussion of the material covered in each lecture. In some cases, they may provide more detail. Exams or quizzes are also designed to determine mastery of the same core concepts covered in the handouts. Note that exams are designed to be comprehensive and will contain material from previous material not necessarily covered in that unit.

b) Online PowerPoints:

- In many instances lecture material will be presented as pre-recorded PowerPoint presentations with voice narration. The pre-recorded lectures will be made available online using the Canvas learning management system. Most topics have related readings in the Mims text that can supplement the material being presented.

c) Turning Point real time questions:

Many lectures will include questions in Turning Point. These questions are intended to foster discussion of the topic and test understanding of the subject. Turning Point questions make up part of the "participation" grade in the course. Participation in class discussions is expected and can be used to gather bonus points for each unit exam grade. Please note: Participation is intended to be "in class" whether in person or by Zoom. **It is NOT acceptable to answer questions for another classmate or provide information to others during the questions.**

Anyone engaging in such activities risks an honor violation.

d) Quizzes:

Quizzes will be given throughout the course. Typically, quizzes will be delivered via Canvas and contain ten questions to test the mastery of specific topics.

1. ***b) Textbook:***

Mims' Medical Microbiology 6th ed. Goering, Dockrell, Zuckerman, Wakelin, and Chiodini. The course schedule lists the chapters or pages in Mims that covers the material discussed in the corresponding lectures.

1. ***c) Additional suggested textbooks and materials***

available in the library

Bhushan, Le, and Ozturk, "FIRST AID for the USMLE STEP 1", 2018 (or latest ed.) McGraw Hill

Gladwin and Trattler, "Clinical Microbiology made ridiculously simple", 3rd edition

Two sets of flashcards for review are also available.

These texts are intended to be review books. They can help you identify the fundamental concepts, but will NOT provide you with enough detail to fully master those concepts.

Medical Microbiology course on the web

Information on the course such as announcements, handouts, clinical cases, and grade reports is also available through the Canvas (Course Info) section. Assignments and links to other sites of interest are also posted to Canvas. PowerPoints. Each student will need to get a password to access the Course Info. If you need instructions on how to access the web check with Dr. Powell (ext. 1582) or Educational Technology.

Use of the handouts and/or materials posted in the intranet is limited to student's use. Duplicating, changing the content or selling those materials is strictly prohibited

Examinations and Grades

There will be six (6) unit exams. Each exam will include a number of questions from each lecture/discussion in that block. Some blocks will contain more content than others and exams will be weighted to reflect how many questions are given in each exam. Each video/lecture plus discussion will represent approximately ten on each of the exams. All questions will be multiple response type questions. You may expect a number of questions in the format of "clinical vignettes" or miniboard-type format. Up to 20% of the questions will be cumulative (they will include material from previous sections). The Microbiology and Immunology Miniboard Exam, provided by the National Medical Board of Examiners (NMBE) will be administered as a comprehensive final exam.

To reward excellent performance on the miniboard exam, bonus points will be awarded on the following basis: Students scoring between 50-74 percentile will receive 1 point to their converted Miniboard score, 75-84th percentile 2 points; 85-89th percentile, 3 points; 90-94th percentile, 4 points, and > 95th percentile, 5 points.

There are six scheduled “Quizzes” throughout the course (see the final Microbiology Schedule for dates). Quiz assignments will vary in format, but will typically be focused on the unit in which they are given. The sixth quiz is optional and may be taken as a makeup for a missed quiz or to drop the lowest quiz grade. In some cases, quizzes will also be included in the video presentations. These are meant as a check to be sure you have understood the information being presented. These are for information alone and will not make up a part of your grade.

A Class participation grade will be given. During the year, we will include questions during the lectures that will be tracked using the Turning Point real-time feedback system. The cumulative Turning Point total will be used to calculate a participation score. Turning Point scores will also be used in determinations of whether to round a grade from a 69 for pass, or 87 for honors to, Turning point scores below 70% will NOT be considered for rounding up. The idea here is to reward those who regularly attend class and answer questions.

Class participation makes up 10% of the final grade. 5% quizzes and 5% Turning Point.

Missed exam policy: Unit exams can be made up **ONLY** in cases where a valid excused absence from the exam is given by the Dean’s office. Preferably this approval should be in advance. However, exceptions can be made under unusual circumstances. Any/all missed unit exams will be made up by taking a comprehensive final examination at the end of the school year. In cases where more than one unit exam is missed each missing grade will be replaced by the student’s score on the makeup exam.

Questions regarding individual unit exams should be brought up with the course director as soon as possible after the exam. If you wish to contest a particular question on an exam you must do so within ten working days. Exam grades will be considered final after that time.

Your final grade in the course will be determined from your performance on all the modes of evaluation, weighted as shown below:

Unit Exams	70%
Miniboard (Final exam)	20%
Turning point and Quiz participation grade	10% quiz TP averages of 70% or greater will receive bonus of 5pts to final average

Pass/Fail/Honors grading. Numerical grades will be reported in Canvas and reported to the registrar as Pass/Fail/Honors based on the following scale:

Fail <70

Pass 70 – 87

Honors 88 - 100

In situations where a student's grade is one point from considerations for rounding to Pass or Honors grade will be based on the overall performance, quizzes, participation grade, unit exams and miniboard. NO extra credit assignments will be given to individual students for the sole purpose of rounding grades to the next letter grade.

Course Remediation: In situations where the student receives a grade below 70 remediation can be done. With consultation from the course director, the student will review video lectures in areas of deficit. The student can show mastery by scoring above 70 on the comprehensive final exam in microbiology. In such cases the student's grade will be reported as: C/D for a successful remediation.

Schedule

Medical Microbiology and Immunology 2023-2024

Red color denotes online video lecture Blue indicates an in-class session (or Live Zoom)

Topic	**Date**	**Day**	**Time**	**Lecturer**
Introduction to the course ()	7-Sep	Thur	9:00 - 10:30	Powell
The organisms ()	7-Sep	Thur	4:00 - 5:00	Powell
Discussion ()	8-Sep	Fri	9:00 - 10:30	Powell
Microbial genetics ()	13-Sep	Wed	1:30 – 3:00	Powell
Discussion ()	18-Sep	Mon	9:00 - 10:30	Powell
Microbial pathogenesis ()	20-Sep	Wed	1:30 – 3:00	Powell
Discussion ()	21-Sep	Thur	9:00 – 10:30	Powell
Lab testing, ()	28-Sep	Wed	1:30 - 2:30	Powell
Discussion ()	29-Sep	Fri	10:30 - 12:30	Powell
Exam #1	**3-Oct**	**Tues**	**1:30**	
Immunology I Intro ()	5-Oct	Thur	1:30 - 3:30	Johnson

Immunology II Vaccines () Immune disorders()	6-Oct	Fri	8:30 - 10:30	Graham
Immunology (extra slot)	6-Oct	Fri	1:30 – 3:30	Johnson
Immunology III Discussion Immunology ()	9-Oct	Mon	8:30 - 10:30	Johnson
Antivirals & susceptibility Testing()	9-Oct	Mon	1:30 - 3:30	Powell
STDs ()	10-Oct	Tue	4:00 – 5:00	Powell
Discussion Antivirals ()	11-Oct	Wed	9:00 – 10:30	Powell
STDs Discussion ()	12-Oct	Thur	9:00 - 10:30	Powell
HIV	16-Oct	Mon	4:00 – 5:00	Johnson
HIV Discussion ()	18-Oct	Wed	8:30 – 10:30	Johnson
Fungal infections ()	19-Oct	Thu	1:30 - 3:00	Powell
Discussion ()	20-Oct	Fri	9:00 - 10:30	Powell
Vector-borne infections ()	20-Oct	Fri	4:00 – 5:00	Powell/Stiles
Discussion ()	23-Oct	Mon	8:30 – 10:30	Powell
Exam #2	**31-Oct**	**Tue**	**10:30**	
Blood borne-Cardiac () Parasites ()	2-Nov	Thu	4:30 - 5:00	Powell/Stiles
Discussion ()	3-Nov	Fri	10:30 - 12:30	Powell

Thanksgiving Holidays	11/22 to 11/24			
UTI ( )	4-Dec	Mon	1:30 - 2:30	Powell
Discussion ()	5-Dec	Tue	10:30 - 12:30	Powell
Exam #3	**19-Dec**	**Tue**	**1:30**	
Winter break	12/21 to 1/1			
Respiratory infections ()	3-Jan	Wed	10:30 - 11:30	Graham?
Discussion ()	3-Jan	Wed	11:30 - 12:30	Graham?
Exam #4	**2-Feb**	**Fri**	**9:30**	
Gastrointestinal infections bacterial/viral () parasites ()	12-Feb	Mon	4:30 - 5:30	Powell/Stiles
Discussion ()	13-Feb	Tues	9:00 - 10:30	Powell/Stiles
Hepatitis ()	14-Feb	Wed	12:30 -1:30	Powell
Discussion ()	16-Feb	Fri	10:30 - 12:30	Powell
Exam #5	**14-Mar**	**Thur**	**1:30**	
Spring Break	3/18 to 3/22			
Pathogen Parade Bacteria ()	25-Mar	Mon	9:00 - 10:30	Powell
Pathogen Parade Virues ()	26-Mar	Tues	9:00 - 10:30	Powell
CNS infections ()	1-Apr	Mon	12:30 - 1:30	Powell/Stiles

Discussion ()	2-Apr	Wed	10:30 - 12:30	Powell/Stiles
Pathogen Paraded for Fungi and Parasites ()	4-Apr	Thur	9:00 - 10:30	Powell/Stiles
Skin infections ()	16-Apr	Tues	1:30 - 2:30	Powell
Discussion ()	17-Apr	Wed	9:00 - 10:30	Powell
Exam #6	**24-Apr**	**Wed**	**9:00**	
Miniboard	**8-May**	**Wed**	**9:00**	
Makeup Exams	**5/13 to 5/17**			

Course Summary:

Date	Details	Due
Wed Aug 15, 2018	Micro (Introduction) (https://msm.instructure.com/calendar?event_id=52577&include_contexts=course_7083)	10:30am to 12:30pm
Wed Aug 22, 2018	Micro (Viruses) (https://msm.instructure.com/calendar?event_id=52582&include_contexts=course_7083)	8:30am to 10:30am
	Micro (Bacteria) (https://msm.instructure.com/calendar?event_id=52598&include_contexts=course_7083)	10:30am to 12:30pm
Fri Aug 24, 2018	Micro (discussion viruses/bac) (https://msm.instructure.com/calendar?event_id=52579&include_contexts=course_7083)	10:30am to 12:30pm
Wed Aug 29, 2018	Micro (Intro to Parasit and Fungi) (https://msm.instructure.com/calendar?event_id=52599&include_contexts=course_7083)	8:30am to 10:30am
Fri Aug 31, 2018	Micro (discussion Parasit/Fungi (https://msm.instructure.com/calendar?event_id=52590&include_contexts=course_7083)	10:30am to 12:30pm
Mon Sep 3, 2018	Labor Day (https://msm.instructure.com/calendar?event_id=52581&include_contexts=course_7083)	12am
Wed Sep 5, 2018	Micro (Lab detection) (https://msm.instructure.com/calendar?event_id=52589&include_contexts=course_7083)	8:30am to 10:30am
Mon Sep 10, 2018	Micro (Lab detection) (https://msm.instructure.com/calendar?event_id=52591&include_contexts=course_7083)	10:30am to 12:30pm

Date	Details	Due
Wed Sep 12, 2018	Micro (Lab discussion) (https://msm.instructure.com/calendar?event_id=52595&include_contexts=course_7083)	10:30am to 12:30pm
Mon Sep 17, 2018	Exam #1 (https://msm.instructure.com/calendar?event_id=52600&include_contexts=course_7083)	9:30am to 10:30am
Mon Sep 24, 2018	Micro (Antivirals) (https://msm.instructure.com/calendar?event_id=52597&include_contexts=course_7083)	10:30am to 12:30pm
Tue Sep 25, 2018	Micro (Antivirals discussion) (https://msm.instructure.com/calendar?event_id=52578&include_contexts=course_7083)	10:30am to 12:30pm
Wed Oct 3, 2018	Micro (Pathogenesis) (https://msm.instructure.com/calendar?event_id=52601&include_contexts=course_7083)	8:30am to 10:30am
	Micro (discussion Pathogenesis) (https://msm.instructure.com/calendar?event_id=52586&include_contexts=course_7083)	10:30am to 12:30pm
Mon Oct 8, 2018	Micro (Normal Flora) (https://msm.instructure.com/calendar?event_id=52583&include_contexts=course_7083)	10:30am to 12:30pm
Wed Oct 10, 2018	Micro (discussion Normal Flora) (https://msm.instructure.com/calendar?event_id=52592&include_contexts=course_7083)	10:30am to 12:30pm
Thu Oct 18, 2018	Exam #2 (https://msm.instructure.com/calendar?event_id=52593&include_contexts=course_7083)	10:30am to 11:30am
Mon Oct 22, 2018	Micro (UTI) (https://msm.instructure.com/calendar?event_id=52580&include_contexts=course_7083)	10:30am to 12:30pm
Wed Oct 24, 2018	Micro (discussion UTI) (https://msm.instructure.com/calendar?	10:30am to 12:30pm

Date	Details	Due
	event_id=52584&include_contexts=course_7083)	
Mon Oct 29, 2018	Micro (Sepsis) (https://msm.instructure.com/calendar?event_id=52585&include_contexts=course_7083)	10:30am to 12:30pm
Wed Oct 31, 2018	Micro (discussion Sepsis) (https://msm.instructure.com/calendar?event_id=52596&include_contexts=course_7083)	10:30am to 12:30pm
Mon Nov 5, 2018	Micro (Fungal) (https://msm.instructure.com/calendar?event_id=52588&include_contexts=course_7083)	10:30am to 12:30pm
Wed Nov 7, 2018	Micro (discussion Fungal) (https://msm.instructure.com/calendar?event_id=52587&include_contexts=course_7083)	10:30am to 12:30pm
Wed Nov 14, 2018	Exam #3 (https://msm.instructure.com/calendar?event_id=52594&include_contexts=course_7083)	9:30am to 10:30am
Fri Nov 6, 2020	Quiz #2 STIs (https://msm.instructure.com/courses/7083/assignments/81903)	due by 11:59pm
Sun Feb 19, 2023	Quiz 2 Gastrointestinal parasites (https://msm.instructure.com/courses/7083/assignments/81900)	due by 11:59pm
Sun Feb 18, 2024	Quiz #3 Hepatitis (https://msm.instructure.com/courses/7083/assignments/81892)	due by 11:59pm
Mon Apr 1, 2024	Quiz #4 (CNS) (https://msm.instructure.com/courses/7083/assignments/81896)	due by 11:59pm
Wed Apr 24, 2024	Quiz (Skin) (https://msm.instructure.com/courses/7083/assignments/81894)	due by 11:59pm
Sun Apr 28, 2024	Quiz Basic Principles (https://msm.instructure.com/courses/7083/assignments/81895)	due by 11:59pm

Date	Details	Due
Wed Aug 28, 2024	Quiz #1 Introductory (https://msm.instructure.com/courses/7083/assignments/86690)	due by 11:59pm
Wed Dec 4, 2024	Quiz 2 (UTI) (https://msm.instructure.com/courses/7083/assignments/81901)	due by 11:59pm
Sun Jan 19, 2025	Quiz 3 Respiratory (https://msm.instructure.com/courses/7083/assignments/81899)	due by 11:59pm
	1-6-2025 10-14 AM (https://msm.instructure.com/courses/7083/assignments/91089)	
	10-17-2024 makeup (https://msm.instructure.com/courses/7083/assignments/87873)	
	10-18-2024 9-43 AM (https://msm.instructure.com/courses/7083/assignments/87874)	
	10-21-2024 10-27 AM (https://msm.instructure.com/courses/7083/assignments/87917)	
	10-21-2024 9-12 AM (https://msm.instructure.com/courses/7083/assignments/87916)	
	10-4-2024 10-24 AM (https://msm.instructure.com/courses/7083/assignments/87647)	
	11-5-2024 9-48 AM (https://msm.instructure.com/courses/7083/assignments/88457)	
	12-5-2024 10-11 AM (https://msm.instructure.com/courses/7083/assignments/88933)	
	8-16-2024 9-54 AM (https://msm.instructure.com/courses/7083/assignments/86157)	
	8-23-2024 9-53 AM (https://msm.instructure.com/courses/7083/assignments/86689)	
	8-28-2024 10-00 AM (https://msm.instructure.com/courses/7083/assignments/86961)	

Date	Details	Due
	[9-12-2024 9-54 AM (https://msm.instructure.com/courses/7083/assignments/87434)](https://msm.instructure.com/courses/7083/assignments/87434)	
	[9-17-2024 9:00 AM (https://msm.instructure.com/courses/7083/assignments/87449)](https://msm.instructure.com/courses/7083/assignments/87449)	
	[Exam #1 (https://msm.instructure.com/courses/7083/assignments/87171)](https://msm.instructure.com/courses/7083/assignments/87171)	
	[Exam #2 (https://msm.instructure.com/courses/7083/assignments/87649)](https://msm.instructure.com/courses/7083/assignments/87649)	
	[Exam #3 (https://msm.instructure.com/courses/7083/assignments/88456)](https://msm.instructure.com/courses/7083/assignments/88456)	
	[Exam #4 (https://msm.instructure.com/courses/7083/assignments/90752)](https://msm.instructure.com/courses/7083/assignments/90752)	
	[Exam #6 (https://msm.instructure.com/courses/7083/assignments/87694)](https://msm.instructure.com/courses/7083/assignments/87694)	
	[Quiz #2 (https://msm.instructure.com/courses/7083/assignments/81926)](https://msm.instructure.com/courses/7083/assignments/81926)	